

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes Annexure A

Data Interpretation

Code of Conduct:

I, K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mounith

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)		Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships		Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification		Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately		Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications		Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	

Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology		Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	
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Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Structure from Motion

Consistency Two datasets produce:

- Dataset A: accurate depth but requires high-quality images
- Dataset B: inconsistent depth but works on low-quality images

Compare the datasets and evaluate which setup is more practical for real-world deployment. Justify your reasoning.

A. Structure from Motion Consistency – Exact Answer

Dataset A and Dataset B have different strengths. Dataset A provides accurate and consistent depth, but it requires high-quality images. This makes it suitable for applications where precise 3D reconstruction and depth accuracy are important. However, its performance may decrease when images contain noise, blur, poor lighting, or low resolution.

Dataset B is more robust because it can work with low-quality images, making it more suitable for uncontrolled real-world environments. However, its inconsistent depth can reduce the accuracy and reliability of the 3D reconstruction.

In Structure from Motion, reliable image features and feature matching are important for estimating camera motion and depth. Therefore, Dataset A is better in terms of accuracy, while Dataset B is better in terms of robustness and practical flexibility.

For general real-world deployment, Dataset B is more practical because real-world cameras may produce low-quality images and environmental conditions cannot always be controlled. It reduces the need for high-quality cameras and strict image conditions. However, if the application requires highly accurate depth, Dataset A should be selected.

Final Decision:

Dataset B is the better choice for general real-world deployment because of its robustness to low-quality images. Dataset A is preferred for accuracy-critical applications where high-quality images are available.

2. Stereo Vision vs Motion

Estimation Depth estimation results:

- Stereo Vision: accurate but requires dual cameras
- Motion Estimation: less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

A. Stereo Vision vs Motion Estimation – Exact Answer

Stereo Vision uses two cameras to estimate depth by finding the difference between corresponding points in the two images. It generally provides more accurate and reliable depth estimation, but it requires two synchronized cameras, proper calibration, and additional hardware.

Motion Estimation estimates depth from the movement of a single camera across multiple frames. It requires less hardware and is easier to deploy, but its depth estimation is generally less accurate and can be affected by camera motion, image quality, and scenes with little or no movement.

Factor	Stereo Vision	Motion Estimation
Depth accuracy	High	Moderate/Low
Hardware	Two cameras	Single camera
Cost	Higher	Lower
Calibration	Required	Less hardware calibration
Hardware constraints	Less suitable	More suitable
Real-time deployment	Good	Good

Evaluation:

When hardware resources are limited, the main advantage of Motion Estimation is that it requires only one camera, reducing hardware cost, size, and complexity. Although it provides lower depth accuracy than Stereo Vision, it can still be effective for applications where moderate accuracy is acceptable.

Final Decision:

Motion Estimation is preferable under hardware constraints because it requires only a single camera, making the system cheaper, simpler, and easier to deploy. Stereo Vision should be preferred when high depth accuracy is more important than hardware cost and complexity.

3. Model Generalization vs

Overfitting Performance:

Model Training Testing

A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

A. Generalization vs Overfitting – Exact Answer

Model A achieves 98% training accuracy but only 70% testing accuracy, while Model B achieves 90% training accuracy and 88% testing accuracy.

Factor	Model A	Model B
Training Accuracy	98%	90%
Testing Accuracy	70%	88%
Generalization	Poor	Good
Overfitting Risk	High	Low
Deployment Reliability	Low	High

Analysis:

Model A performs extremely well on the training data but its accuracy drops significantly on unseen testing data. The 28% gap between training and testing accuracy indicates overfitting. This means Model A has likely learned the training data too closely and may perform poorly on new real-world data.

Model B has a smaller 2% gap between training and testing accuracy. Its testing accuracy of 88% is also much higher than Model A's 70%. Therefore, Model B demonstrates better generalization and is less likely to fail when exposed to unseen data.

Conclusion:

Model B is more reliable for deployment because it provides high testing accuracy, has a very small training-testing gap, and has a much lower risk of overfitting. Although Model A has higher training accuracy, its poor testing performance makes it less suitable for real-world deployment.

Final Decision: → Model B should be selected for deployment because it generalizes better to unseen data and has lower overfitting risk.

4. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

A. YOLO Performance Across Scales – Exact Answer

YOLO shows different detection performance depending on object size. It achieves 95% accuracy for large objects, 85% for medium objects, and only 60% for small objects.

Object Size	Detection Accuracy	Performance
Large	95%	Excellent
Medium	85%	Good
Small	60%	Poor

Analysis:

YOLO performs very well for large objects because they occupy more pixels in the image, making their features easier to detect. Performance decreases for medium objects and drops significantly for small objects because small objects contain fewer pixels and their important features can be lost during image resizing and feature extraction.

The 35% difference between large and small object detection shows that YOLO alone may not provide sufficiently reliable multi-scale detection, especially when small objects are important.

Required Improvements:

- Use multi-scale feature extraction such as Feature Pyramid Networks (FPN).
- Use higher-resolution input images to preserve small-object details.
- Improve feature fusion between different scales.
- Use appropriate data augmentation and small-object training samples.
- Consider advanced YOLO versions or architectures specifically optimized for small-object detection.

Conclusion:

YOLO alone is not sufficient for highly reliable multi-scale detection when small objects are critical. It performs excellently on large objects but needs additional multi-scale techniques and higher-resolution features to improve the 60% small-object detection accuracy.

Final Decision: → YOLO is effective for large and medium objects, but multi-scale feature extraction and small-object optimization are required for reliable detection across all scales.

5. Motion Model Selection for

Animation Two models:

- Model A: smooth but physically inaccurate
- Model B: realistic but computationally intensive

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

A. Model A and Model B have different advantages. Model A produces smooth motion with lower computational requirements but is physically inaccurate, whereas Model B produces realistic and physically accurate motion but requires more computational resources.

B. Factor	C. Model A	D. Model B
E. Smoothness	F. High	G. High
H. Physical accuracy	I. Low	J. High
K. Realism	L. Moderate	M. High
N. Computational cost	O. Low	P. High
Q. Real-time performance	R. Suitable	S. May be difficult
T. Cinematic quality	U. Limited	V. Excellent

W. Analysis:

Model A is preferable when real-time performance is the main requirement. Its low computational cost allows animation to run smoothly with limited processing power. However, its physically inaccurate motion may reduce realism.

X. Model B is preferable when visual realism and physical accuracy are more important than computational cost. Although it requires more processing power, it can produce realistic motion suitable for high-quality animation.

Y. Conclusion:

For real-time animation systems such as games, interactive applications, and simulations, Model A should be selected because it provides smooth motion with lower computational requirements.

Z. For cinematic animation, movies, and high-quality visual effects, Model B should be selected because realistic and physically accurate motion is more important than computational cost.

AA. Final Decision:

Real-time constraint → Model A

Cinematic/high-realism requirement → Model B